POSTER ID: SUS-PP-18

Enhanced Primary Treatment and Hybrid Wastewater Treatment Systems for Domestic Sewage

Md Sarfraz Ahmad[1], Maithili Mohanty[2], Md Sarfraz Ahmad[1,*]

[1,2]Department of Civil Engineering, National Institute of Technology, Rourkela, Odisha, India-769008

EMAIL 525ce6007@nitrkl.ac.in | * Corresponding author

NIT ROURKELA, ODISHA

SCAN FOR MORE

01 INTRODUCTION

Rapid urbanization and population increase have led to tremendous pressure on residential sewage infrastructure. Conventional treatment is capable of removing only a limited proportion of suspended and organic solids, which is a burden for downstream biology.

To bridge this gap, Chemically Enhanced Primary Treatment (CEPT) was employing coagulants typically ferric chloride, alum, to enhance solids capture, under optimal dosage. However, CEPT is leaving more sludge and residual coagulant [1]. These restrictions have resulted in the transition to hybrid wastewater treatment systems that combine Anaerobic and Aerobic processes or nature-based solutions to tackle the organic loading and suspended particles [2]. This poster assesses chemically enhanced and hybrid treatment techniques for scalable, cost-effective alternatives for domestic sewage with focus on removal efficiency and operational feasibility.

02 OBJECTIVES

- ✓ To analyze and compare the mechanism of CEPT and some hybrid treatment techniques.
- ✓ To review the effectiveness for pollutant removal [BOD, COD, and TSS] using CEPT and some hybrid treatment techniques.

03 METHODOLOGY

A systematic and extensive investigation of literature was conducted for this review to evaluate the present status, performance and future outlook of Enhanced Primary Treatment and Hybrid anaerobic-aerobic treatment systems for domestic sewage. The methodology of the review involved determining the scope and objectives of the research. A thorough search was carried out in the major scientific databases such as Scopus, Web of Science, ScienceDirect, SpringerLink, Google Scholar and PubMed, with a primary focus on peer-reviewed publications published between 2010 and 2026. The search strategy was then carefully developed utilizing relevant keywords and Boolean operators and depicted in flow chart.

RESEARCH QUESTION
Domestic sewage treatment, conventional treatment, move to Enhanced primary treatment, Anaerobic and Aerobic digestion

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DATA BASES & SEARCH ENGINES
Google Scholar, Web of Science, Google, Research Gate, Scopus, Science Direct, McGraw hill between Literature Search (2010-2026)

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SEARCHES USED COMBINATIONS OF BOOLEAN OPERATORS
Domestic sewage and conventional treatment, Sewage treatment by EPT/CEPT, Domestic sewage and hybrid treatment, domestic wastewater and MBBR

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SCREENING OF LITERATURE
Removes the literatures including industrial wastewater, desalination treatment, literatures showing only anaerobic and aerobic treatment

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TECHNOLOGY CLASSIFICATION AND MECHANISM INTERPRETATION
Selection of CEPT and hybrid system and explaining the mechanism and comparing them

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DATA EXTRACTION AND PERFORMANCE ANALYSIS
Create graphs and datasheets using key parameters TSS, BOD, COD, Biogas yield in Anaerobic and Aerobic processes, compare efficiencies, energy demand, cost, land footprint

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KNOWLEDGE GAP
Identify limitations and future research opportunities

Figure 1. Schematic of the proposed methodology.

04 RESULTS

The mechanism shows that CEPT eliminates suspended particles and organic materials through the use of coagulants. In contrast, the hybrid treatment system integrates the suspended solids and organic matter through anaerobic and aerobic processes.

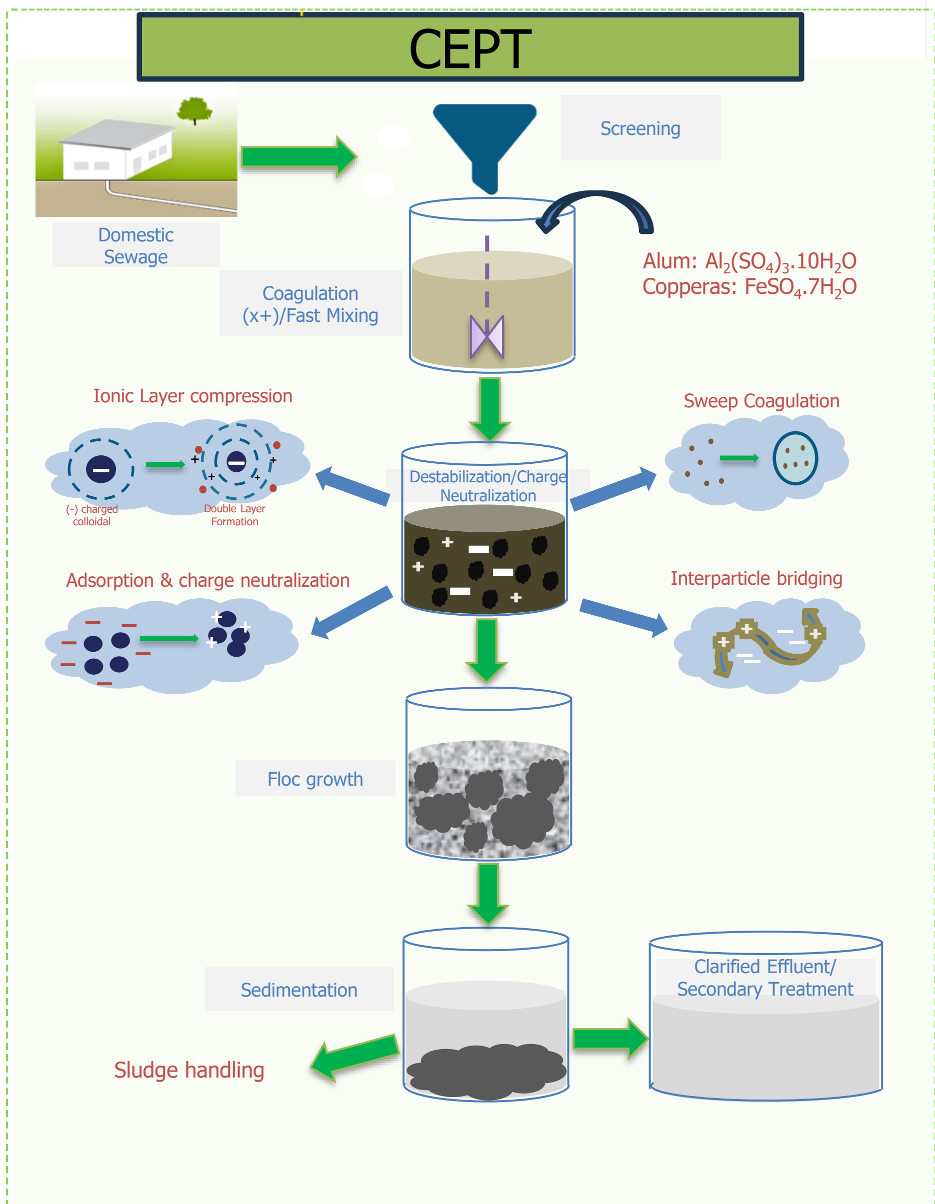


Figure 2. Mechanism process of CEPT.

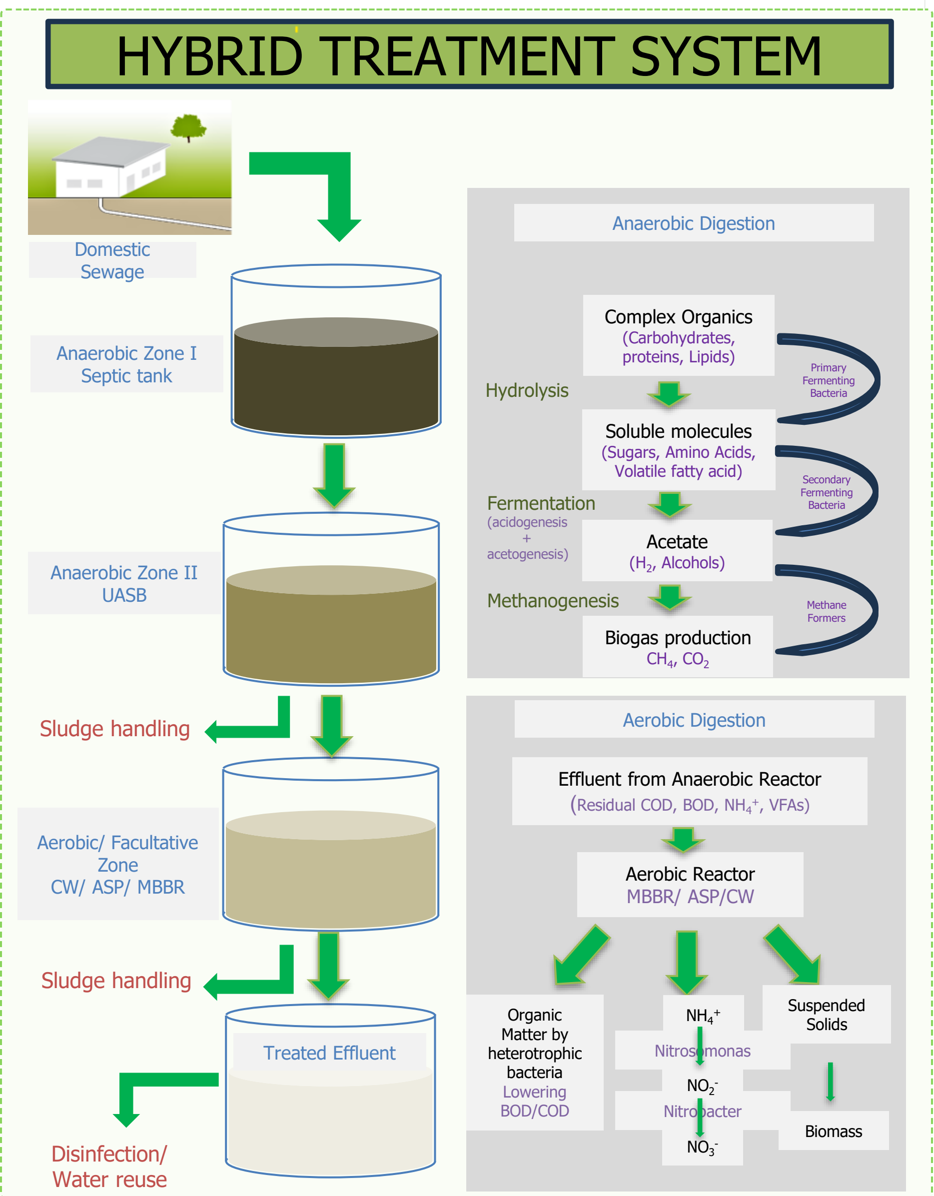


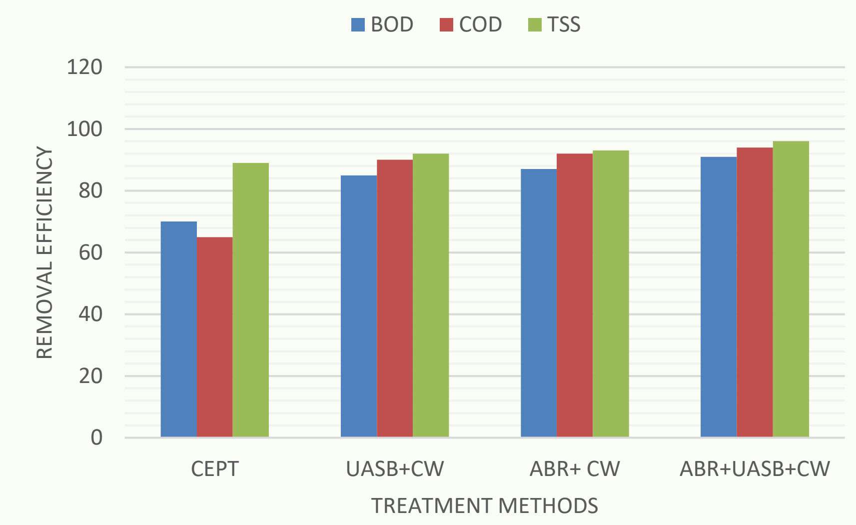
Figure 3. Mechanism process of Hybrid Treatment System.

05 DISCUSSION

The comparative study of CEPT and hybrid treatment systems for the removal efficiency of BOD, COD, and TSS, alongside operational costs and biogas recovery, is presented in the table and illustrated with graphs.

Table 1. Comparison against CEPT and hybrid treatment system.

PARAMETER	CEPT(eff. %)	HYBRID(eff. %)	Ref(s)
BOD(Biochemical Oxygen Demand)	70-80	85-93	-
COD(Biochemical Oxygen Demand)	60-70	80-90	[3]
TSS(Total Suspended Solids)	89-95	90-94	[4]
Operational cost	High	Moderate	
Biogas recovery	No	Yes	
Energy requirement	Low	Moderate	
Land footprint	Moderate	Low-moderate	



Graph 1. Removal efficiency of different treatment technology

ABBREVIATIONS
EPT- Enhanced primary treatment
CEPT- Chemically Enhanced primary Treatment
UASB- Up flow Anaerobic Blanket Reactor
ABR- Anaerobic Baffled reactor
CW- Constructed Wetland

06 CONCLUSIONS

- CEPT is a pretreatment technology that effectively removes suspended solids and a significant portion of organic matter however reliance on chemical dosing and generate a lot sludge.
- Hybrid treatment systems result in higher removal of BOD, COD, and TSS as well as biogas recovery and better effluent quality for water reuse.
- The literature suggests that hybrid treatment systems provide a higher level of sustainability through less sludge generation, lower long-term running costs, energy recovery, and lower land footprint.

"*Wastewater is a valuable resource in a world where water is finite and demand is growing*"
— GUY RIDER

07 FUTURE WORK

Future research should focus on optimizing the integration of CEPT with anaerobic and aerobic processes to improve energy recovery, nutrient removal and operational stability while minimizing chemical use and sludge creation.

08 REFERENCES

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09 ACKNOWLEDGEMENTS

This work is supported and funded by ANRF [Anusandhan National Research Foundation], Grant no. [ANRF/ECRG/2024/000351/EN].